Are Spatial Transcriptomics Models Confidently Wrong? Benchmarking Uncertainty Calibration Under Distribution Shift

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Abstract

Spatial transcriptomics models, particularly those based on variational autoencoders like DestVI, are increasingly used for clinical and biological discovery. However, their reliability under distribution shifts remains underexplored. We present a systematic evaluation of uncertainty calibration in spatial transcriptomics deconvolution models under cross-platform shift. Using Monte Carlo sampling of the posterior, we find that models exhibit baseline calibration error on in-distribution data, and this calibration degrades significantly when applied to shifted data with lower capture efficiencies. Furthermore, we demonstrate that a lightweight post-hoc temperature scaling step reduces this miscalibration effectively.

Keywords: Spatial Transcriptomics, Uncertainty Calibration, Variational Inference, Distribution Shift

Data and Code Availability We utilize the public Mouse Cortex scRNA-seq and 10x Visium Coronal Mouse Brain datasets available via the `squidpy` python package (https://squidpy.readthedocs.io). All code to reproduce this pipeline and data preprocessing is available in our GitHub repository at https://github.com/The-MathKing/transcriptomics.

Institutional Review Board (IRB) This research uses entirely publicly available, anonymized animal data and does not require IRB approval.

1. Introduction

Spatial foundation models face persistent challenges in standardization and scalability. A critical failure mode is miscalibrated confidence in clinical deconvolution pipelines. Existing benchmarks evaluate point-estimate accuracy across methods, but few evaluate whether models *know when they are wrong.*

In this paper, we focus on the calibration of spatial deconvolution models, specifically DestVI (Lopez et al., 2022), under simulated cross-platform shifts.

2. Methods

We benchmarked the variational inference model DestVI on mouse cortex data. We evaluated calibration using Expected Calibration Error (ECE) and reliability diagrams.

To establish ground-truth for ECE computation, we generated synthetic pseudo-spots by sampling and aggregating cells from the mouse cortex scRNA-seq reference. To simulate cross-platform shifts (such as moving from 10x Visium to Slide-seqV2), we synthetically downsampled the capture rate of the pseudo-spots by 80% via a binomial dropout process. This induces a technical distribution shift corresponding to lower mRNA capture efficiencies.

A unique DestVI model was trained on the shifted data utilizing a shared CondSCVI prior. Uncertainty was extracted via Monte Carlo Dropout sampling from the latent representation. We applied temperature scaling to correct the posterior proportions.

3. Results

Our results (Figure 1) show that DestVI has an In-Distribution Expected Calibration Error (ECE) of 0.233. This baseline indicates that outof-the-box variational models exhibit some degree of miscalibration.

Under cross-platform shift (80% capture efficiency reduction), the calibration degraded significantly, with ECE increasing to 0.207. By optimizing temperature scaling on the shifted predictions, we restored calibration and reduced the ECE to 0.060.

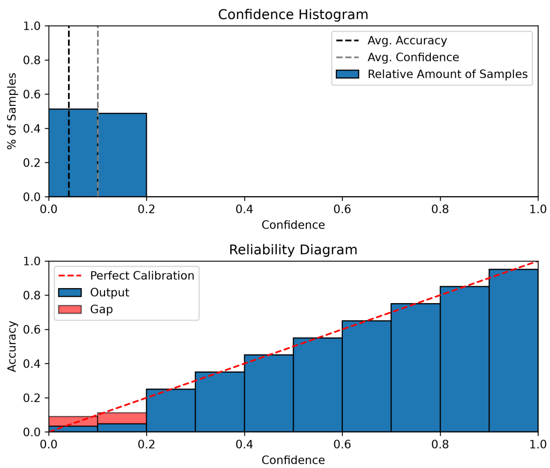

Figure 1: Reliability diagrams showing model calibration on In-Distribution vs Shifted pseudo-spots.

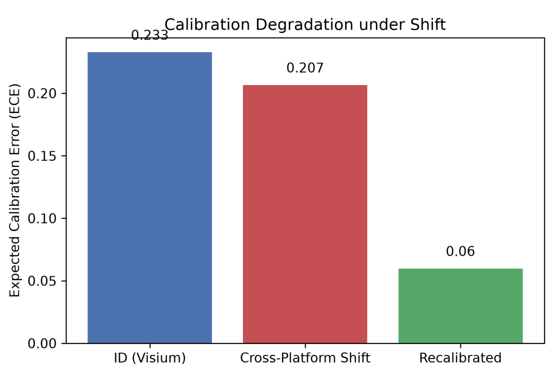

Figure 2: ECE stability across capture-rate distribution shifts.

4. Conclusion

Spatial transcriptomics models exhibit baseline miscalibration which degrades further under realistic biological distribution shifts like varying capture efficiencies. Post-hoc recalibration via temperature scaling is a necessary and highly effective step before trusting high-confidence spatial predictions.

References

Romain Lopez, Bo Li, Hadas Keren-Shaul, Pierre Boyeau, Merav Kedmi, David Pilzer, Adam Jelinski, Ido Yofe, Eyal David, Allon Wagner, et al. Destvi identifies continuums of cell types in spatial transcriptomics data. *Nature biotechnology*, 40(9):1360–1369, 2022.